

Ultra Fast Copy

User Manual — high-performance file copy/move for Windows (CLI + GUI)

Ultra Fast Copy targets the cases where Windows Explorer bogs down: hundreds of thousands of files, network drives, and interrupted jobs. It adapts its parallelism to file size, storage device, and network conditions, and adds verification, retry, and resume on top of a shared core engine used by both the command-line tool (ufCopy) and the graphical app (ufCopyTool).

1. Installation

Easiest: run UltraFastCopy-Setup.exe. It installs both executables for the current user (no administrator rights needed) and adds the **"Open with Ultra Fast Copy"** entry to the Explorer right-click menu. To remove everything, run UltraFastCopy-Setup.exe /uninstall.

Otherwise, download the standalone executables from the release page, or build from source:

```
uv venv --python 3.12
uv pip install -e ".[dev]"
# or, without uv:
py -3.12 -m venv .venv
.venv\Scripts\activate
pip install -e ".[dev]"
```

Two executables are provided:

File	What it is	Qt bundled
ufCopy.exe	Command-line interface	No (permissive-only)
ufCopyTool.exe	Graphical interface	PySide6 (LGPL-3.0)

2. Command-line usage

```
ufCopy copy "D:\Source" "E:\Backup" --workers 12 --verify xxhash --conflict overwrite-if-newer --prescan
ufCopy move "D:\Source" "E:\Archive" --verify xxhash --retry 5
ufCopy jobs
ufCopy resume <JOB_ID>
ufCopy benchmark "D:\Source" "E:\Temp"
ufCopy config show
ufCopy config init
ufCopy gui
```

Key options

Option	Description
--workers N	Worker thread count. Automatic if omitted.
--buffer-size 8MiB	Copy buffer size.
--verify MODE	none size mtime_size xxhash sha256
--conflict POLICY	What to do when the destination exists.
--retry N	Retries for transient errors (exponential backoff).
--prescan / --streaming	Count everything first / start while scanning.
--include / --exclude	glob patterns (repeatable).
--dry-run	Report the plan without writing.
--json-output	Emit JSON events to stdout for automation.
--bandwidth-limit 10MiB	Cap transfer rate per second.
--preset fast balanced safe	Speed preset.

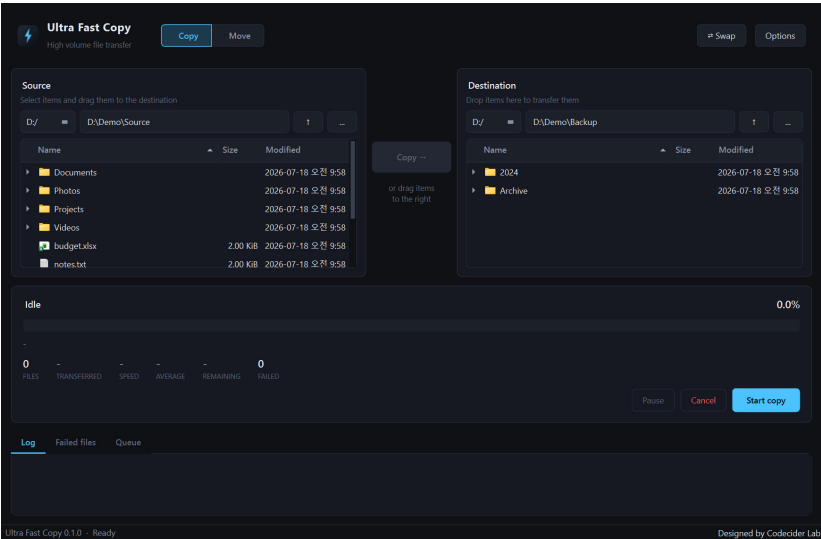
Exit codes

Code	Meaning
0	Success
1	Completed, but some files failed
2	Job failed
3	Cancelled by the user
4	Invalid usage

3. Graphical app

ufCopyTool

```
ufCopyTool --source "D:\Source" --destination "E:\Backup"
```



The main window: source tree (left), transfer controls (center), destination tree (right), progress panel, and Log / Failed files / Queue tabs.

Running a transfer

1. Pick the source folder in the **left** pane and the target in the **right** pane (use the drive selector, the path box, or the "..." button).
2. Choose **Copy** (default) or **Move** with the toggle at the top. Move asks for confirmation before it deletes anything.
3. Select one or more items in the source pane (Ctrl/Shift click for several).
4. Start the transfer either by pressing **Start copy**, by clicking the center **Copy** → button, or by **dragging** the selection from the left pane onto the right pane.
5. Watch progress, speed, average speed, ETA, and the failed count in the panel. **Pause**, **Resume**, or **Cancel** at any time.

Other controls

- **Swap** exchanges the source and destination panes.
- **Options** opens the detailed settings (verification mode, conflict policy, worker count, buffer size, speed preset, filters).
- The **Log** tab shows messages, **Failed files** lists per-file errors, and **Queue** shows pending jobs.
- Dark theme, code-drawn icon. Date and size formatting follow the Windows locale.

4. Explorer right-click menu

```
ufCopy shell install      # register for the current user (no admin needed)
ufCopy shell status      # show registration state
ufCopy shell uninstall    # remove
```

Adds an **"Open with Ultra Fast Copy"** entry for files, folders, folder backgrounds, and drives. On Windows 11 registry-based entries appear under **"Show more options"** (Shift+F10).

5. Configuration

Settings live in %APPDATA%\UltraFastCopy\config.toml; CLI arguments take precedence. Logs are written to %LOCALAPPDATA%\UltraFastCopy\logs\ and resume checkpoints to %LOCALAPPDATA%\UltraFastCopy\checkpoints\.

6. Performance

Ultra Fast Copy does not promise to beat Explorer or Robocopy everywhere. Its value is low UI overhead on many-file jobs, environment-aware concurrency, and data integrity through verification, retry, resume, and detailed logs. Measure your own environment with `ufCopy benchmark`. A detailed study of how to beat Robocopy on NTFS (directory-lock scatter, block-level imaging) ships in `docs/benchmark.md`.

7. License

Ultra Fast Copy is released under the MIT License. The GUI executable bundles PySide6 (LGPL-3.0); the CLI executable contains only permissive-licensed components. See `LICENSE` and `THIRD_PARTY_LICENSES.md`.

Designed by Codecider Lab · Ultra Fast Copy 0.1.0